Class Imbalance in Histopathology: Prevalence Alone Costs Little Accuracy on TCGA-UT

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Abstract

In a long-tailed training set, the rare classes usually receive both a smaller share of the patches and fewer patients. This report isolates the first factor, class prevalence, at fixed patient support. On TCGA-UT, a multinomial logistic classifier on frozen Virchow2 features is trained on the same twenty patients per cancer type and the same total of 19,200 patches in every arm; only the share of the budget that each of the 30 classes receives changes, following an exponential long-tail profile with imbalance ratio ρ from 1 to 100 and a random class order per draw. A further arm uses the classes' native training shares, which realize $\rho = 18.1$. Balanced accuracy degrades slowly and with increasing speed: it falls by 0.57 percentage points at $\rho = 10$ (95% interval: 0.30 to 0.84), by 0.996 points at $\rho = 20$ (0.66 to 1.34), just short of the one-point threshold, by 1.80 points at $\rho = 50$ (1.41 to 2.19), the first ratio that meets it, and by 2.76 points at $\rho = 100$ (2.34 to 3.20). The loss per doubling of ρ is 0.17 points up to $\rho = 10$ and 0.65 points beyond. Damage concentrates on the tail third of the classes, whose recall drops by 11.5 points at $\rho = 100$ while the head third gains 5.3. The native allocation costs 1.17 points (0.84 to 1.47) and lies on the ratio curve at its realized ratio. Probability quality deteriorates much more than accuracy: at $\rho = 100$ the expected calibration error rises from 5.5 to 18.6 points. A single validation temperature removes 95 percent of that increase and 77 percent of the increase in macro negative log-likelihood. On TCGA-UT, prevalence at fixed patient support is therefore a moderate source of discrimination loss, and its calibration damage is mostly a matter of scale.

1 Motivation

The benchmark of this series found that severe class imbalance damages balanced accuracy on TCGA-UT by 1.33 to 1.79 percentage points, and that validation temperature scaling removes most of the accompanying differences in probability quality (Queisler, 2026b,a). These numbers bundle several factors. Each imbalanced condition changed the prevalence of the classes and the patients available to them at the same time, it was measured at three allocations only (balanced, moderate, and severe, $\rho \approx 100$), and the classifier was a two-layer perceptron trained alongside five mitigation methods. A later comparison showed that multinomial logistic regression on the same features differs from that perceptron by at most half a point (Queisler, 2026d), and all subsequent studies of this series use it.

This report begins a set of small studies that separate the deprivations of the benchmark, one factor at a time and with the plain cross-entropy baseline only. It isolates class prevalence. The question is: *how fast do balanced accuracy, negative log-likelihood, and calibration error degrade as the training class distribution moves from balanced to long-tailed, when the patients of every class and the total patch budget stay fixed, and where does the dataset's native distribution lie on that curve?* Patient shortage and patch shortage per class are held out of this design and left to their own studies.

2 Design

The study uses the 30 TCGA-UT cancer types with frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), the three fixed patient-disjoint splits of the benchmark, and ten draws per split. Validation and test partitions keep their natural class distribution. In each draw, twenty patients per class are sampled from those with at least $D = 160$ training patches, and the same patients enter every arm of the draw. Each patient's patches follow one fixed round-robin order over slides, and an arm that assigns m patches to a patient takes the first m of them. Arms of the same draw therefore differ only in how deep into each patient's ordering they read, and a class's smaller allocation is always contained in its larger one.

Every arm trains on the same total budget $T = K \cdot G \cdot 32 = 19{,}200$ patches for $K = 30$ classes and $G = 20$ patients, which is 32 patches per patient in the balanced arm. A ratio arm with imbalance ratio ρ assigns the class at rank $k \in \{0, \ldots, K-1\}$ the count

$$N_k = N_0 \, \rho^{-k/(K-1)}, \tag{1}$$

with N_0 chosen so that the counts sum to T, rounded to integers that keep the sum, at least one patch per patient, and at most D per patient. This exponential profile is the allocator of the benchmark (Queisler, 2026a). It moves the class prior and the tail support together: at $\rho = 100$ the head class receives 2,843 patches, about 142 per patient, and the tail class 28, between one and two per patient. The assignment of classes to ranks is a random permutation drawn per split and draw, so each cancer type visits head, body, and tail across the 30 draws, and class difficulty averages out of the curve. Within a class, the count is spread as evenly as possible over its twenty patients.

The native arm N gives each class its share of all training patches of the split, scaled to T, in the classes' own order and with the same twenty patients. Its counts range from about 95 to about 1,700 patches per class, and its realized ratio, the largest over the smallest class count, averages 18.1 over the 30 draws. Table 1 lists the arms, and Figure 1 shows every realized class distribution.

Arm	Realized ρ	Head class	Tail class	Tail per patient
r1	1.0	640	640	32
r2	2.0	886	443	22.1
r5	5.0	1,278	256	12.8
r10	10.0	1,615	161	8.1
r20	19.9	1,973	99	5.0
r50	50.3	2,466	49	2.5
r100	101.5	2,843	28	1.4
N	18.1	≈1,700	≈95	≈4.8

Table 1: All arms train on the same twenty patients per class and the same 19,200 patches; only the per-class share changes. Head and tail are the largest and smallest class counts in patches. The realized ratio is averaged over 30 split-draw pairs and departs from the nominal ratio only through integer rounding. The native arm N uses the training split's own class shares, so its extreme counts vary slightly between splits.

Each arm fits a multinomial logistic classifier whose regularization strength is selected from nine values between 10^{-6} and 100 by validation patient-macro balanced accuracy (Queisler, 2026d). After selection, one temperature is fitted on the validation logits by minimizing their negative log-likelihood and applied to the test logits (Guo et al., 2017); it rescales confidence and leaves

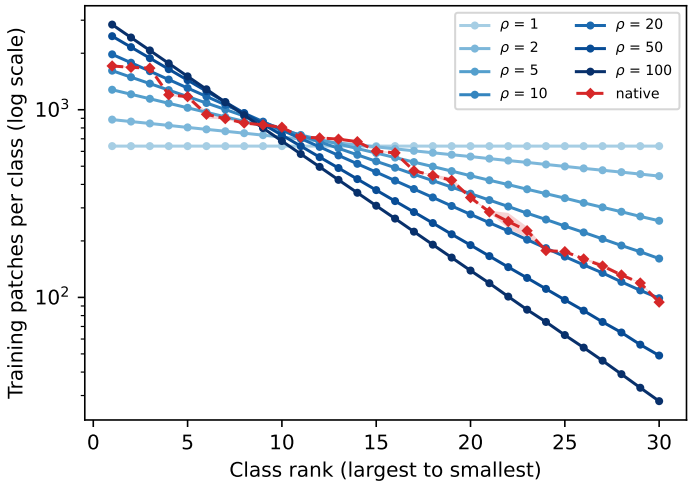


Figure 1: The ratio arms tilt one fixed budget around the middle classes, and the native allocation follows no single exponential profile. Training patches per class, sorted from the largest to the smallest class, on a logarithmic axis, so an exponential profile is a straight line. Blue lines are the ratio arms, from light ($\rho = 1$) to dark ($\rho = 100$); their sorted counts are identical in every draw. The red dashed line is the native arm, averaged over the 30 split-draw pairs, with the shaded band spanning the smallest and largest count at each rank. All arms sum to 19,200 patches.

predictions unchanged. The study fits eight arms for each of 30 split-draw pairs, 240 classifiers in total, and every fit converged. Test data never entered selection or temperature fitting.

3 Evaluation

The primary endpoint is the *damage* $\Delta(\rho) = \mathrm{BA}(1) - \mathrm{BA}(\rho)$, where BA is patient-macro balanced accuracy on the test partition, computed as the mean of per-class patch recalls in which every test patient of a class carries equal weight. Estimates are pooled over splits and draws with equal weight per split. Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split over 2,000 replicates; all arms share the replicates, so every contrast is paired, and intervals are 95 percent percentile intervals. Prevalence counts as damaging at ratio ρ if $\Delta(\rho)$ is at least one percentage point and its interval lies entirely above zero; the *onset* is the smallest ratio of the grid that meets this criterion. The *slope* is the least-squares slope of BA on $\log_2 \rho$ per replicate, in points per doubling of ρ, computed over the whole grid and separately over $\rho \leq 10$ and $\rho \geq 10$.

Probability quality is measured by two secondary endpoints, both with and without temperature scaling. The macro negative log-likelihood (NLL) averages the log-loss of the true class so that every test patient and every class carry equal weight; it depends on discrimination as well as calibration. The expected calibration error (ECE) compares top-label confidence with accuracy in ten fixed confidence bins, with patient weights. Both metrics are known to depend on the class distribution of the evaluation data (Mosquera et al., 2024); here that distribution is the same for all arms. Two diagnostics complete the evaluation. The recall of the head, body, and tail thirds of the classes, by their assigned rank, shows where the damage lands. The native arm is compared both with the balanced arm and with the ratio curve, interpolated linearly in ρ at N's realized ratio.

Three readings were fixed in advance. If Δ stays below one point up to $\rho = 100$, prevalence alone is not a material deficit on this dataset at fixed patient support, and the imbalance damage

of the benchmark must come from the factors bundled with it. If Δ crosses the threshold within the grid, the onset and slope locate the ratio from which a prevalence correction is worthwhile. If the native arm lies below the curve, the native order of the classes, which ties prevalence to specific cancer types, costs more than a random order.

4 Results

The results follow the endpoints: first the accuracy curve with its onset and slope, then the classes on which the damage lands, the position of the native allocation, and finally probability quality before and after temperature scaling. The balanced arm reaches 72.05 percent balanced accuracy (70.68 to 73.31). Twenty patients with eight patches each reached 71.89 percent in Queisler (2026c); the fourfold patch depth of this design therefore changes the balanced reference by less than a fifth of a point, as expected on TCGA-UT, where patch depth hardly matters.

4.1 Accuracy curve

Balanced accuracy falls monotonically with the imbalance ratio, slowly at first and then faster (Table 2, Figure 2). Already $\rho = 2$ costs a resolved 0.13 points (0.03 to 0.23), but the damage stays below half a point up to $\rho = 5$ and reaches 0.57 points (0.30 to 0.84) at $\rho = 10$. At $\rho = 20$ it is 0.996 points (0.66 to 1.34), just short of the one-point threshold. At $\rho = 50$ the damage is 1.80 points (1.41 to 2.19), with the whole interval above the threshold, so the onset of the grid is $\rho = 50$. At $\rho = 100$ the damage is 2.76 points (2.34 to 3.20). The three splits agree closely: the damage at $\rho = 100$ is 2.73, 2.81, and 2.74 points.

			Macro NLL (nats)		ECE (points)		
Arm	BA (%)	Damage Δ	raw	scaled	raw	scaled	Temperature
r1	72.05	—	0.943	0.922	5.52	1.51	0.85
r2	71.92	0.13 [0.03, 0.23]	0.951	0.929	5.56	1.53	0.92
r5	71.70	0.36 [0.13, 0.57]	0.974	0.946	6.32	1.54	1.17
r10	71.48	0.57 [0.30, 0.84]	0.991	0.954	6.80	1.65	1.22
r20	71.06	0.996 [0.66, 1.34]	1.083	0.975	10.27	1.74	1.42
r50	70.25	**1.80** [1.41, 2.19]	1.286	1.016	15.02	1.91	1.78
r100	69.29	**2.76** [2.34, 3.20]	1.543	1.060	18.65	2.11	2.16
N ($\rho = 18.1$)	70.88	**1.17** [0.84, 1.47]	1.105	0.980	9.64	1.40	1.46

Table 2: Accuracy degrades slowly with prevalence, raw probability quality much faster, and temperature scaling removes most of the probability-quality loss. Balanced accuracy (BA) is test patient-macro balanced accuracy; the damage is $\mathrm{BA}(1) - \mathrm{BA}(\rho)$ in percentage points with paired 95% bootstrap intervals, in bold where it meets the one-point criterion. NLL and ECE are shown without (raw) and with (scaled) the validation temperature; the temperature column is the mean fitted temperature over 30 fits, where values above one soften overconfident predictions. The 95% intervals of the absolute BA levels are about 2.6 points wide and are shared by all arms.

The loss per doubling of ρ is 0.39 points (0.33 to 0.46) over the whole grid, but the curve is not linear in $\log \rho$. Up to $\rho = 10$ it loses 0.17 points per doubling (0.08 to 0.26), from $\rho = 10$ to $\rho = 100$ it loses 0.65 points (0.59 to 0.71), and the two intervals do not overlap. The damage thus accelerates as the tail classes approach one or two patches per patient.

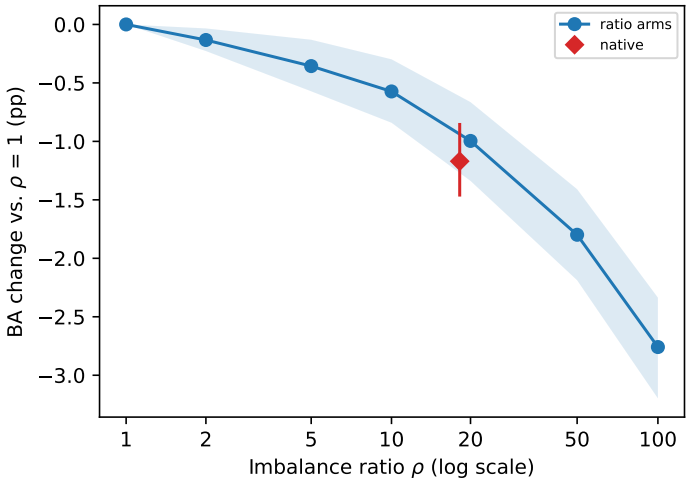


Figure 2: Balanced accuracy degrades slowly and accelerates beyond $\rho = 10$. The line shows the change in patient-macro balanced accuracy relative to the balanced arm, with the ratio arms at their realized ratio and a shaded paired 95% bootstrap band; the red diamond marks the native allocation at its realized ratio of 18.1.

4.2 Where the damage lands

The aggregate damage is the net result of a large redistribution of recall from the tail to the head (Table 3). In the balanced arm the three thirds of the class ranking reach the same recall, about 72 percent, as the random rank assignment implies. As ρ grows, the head third gains up to 5.3 points and the tail third loses up to 11.5 points, while the body third loses 2.1 points at $\rho = 100$. Because balanced accuracy weights all classes equally, the loss of the tail outweighs the gain of the head, and the balance between them worsens with ρ: at $\rho = 10$ the tail loses 5.0 points against a head gain of 4.0, at $\rho = 100$ it loses 11.5 against 5.3. The head gain saturates beyond $\rho = 20$, whereas the tail loss keeps growing, which produces the acceleration of the curve.

Third	r1	r2	r5	r10	r20	r50	r100
Head	72.04	73.87	74.68	76.09	76.60	77.06	77.38
Body	72.18	72.13	71.76	71.46	71.01	70.47	70.05
Tail	71.93	69.76	68.65	66.90	65.56	63.23	60.46

Table 3: Imbalance moves recall from the tail to the head, and the tail loses more than the head gains. Mean patient-macro recall in percent of the ten head, body, and tail classes of each draw, by their assigned rank, averaged over 30 fits. The native arm is omitted because its class order is not the random rank order that defines the thirds.

4.3 Native allocation

The native class distribution of TCGA-UT costs 1.17 points against the balanced arm (0.84 to 1.47), just above the threshold. It lies on the ratio curve: interpolated at its realized ratio of 18.1, the curve predicts a damage of 0.92 points, and the native arm falls 0.25 points below it (-0.58 to 0.06), a difference whose interval contains zero. The native order ties prevalence to particular cancer types, whereas the ratio arms randomize that assignment, so this agreement indicates that the rare cancer types of TCGA-UT are not systematically harder or easier to learn from few patches than the average class. The native profile is also not exponential: its median class receives about 700 patches, whereas an exponential profile with the same extremes gives

the median class about 400. The realized ratio alone nevertheless places it on the curve within the precision of this study.

4.4 Probability quality

Raw probability quality degrades long before accuracy does, and most of the degradation is removed by a single temperature (Table 2, Figure 3). Without scaling, the ECE stays near 5.5 points up to $\rho = 2$ and rises to 10.27 points at $\rho = 20$ and 18.65 points at $\rho = 100$, an increase of 13.12 points (11.35 to 15.02). Macro NLL rises by 0.60 nats (0.51 to 0.71) at $\rho = 100$. After temperature scaling, the increase at $\rho = 100$ shrinks to 0.59 points of ECE (0.26 to 0.95) and 0.14 nats of NLL (0.12 to 0.15). Scaling therefore removes 95 percent of the ECE increase and 77 percent of the NLL increase. The scaled NLL increase that remains is resolved and follows the accuracy loss, since NLL also penalizes wrong predictions. The native arm shows the same pattern: its raw ECE exceeds that of the balanced arm by 4.12 points (2.25 to 6.07), and after scaling the difference is -0.12 points (-0.37 to 0.14).

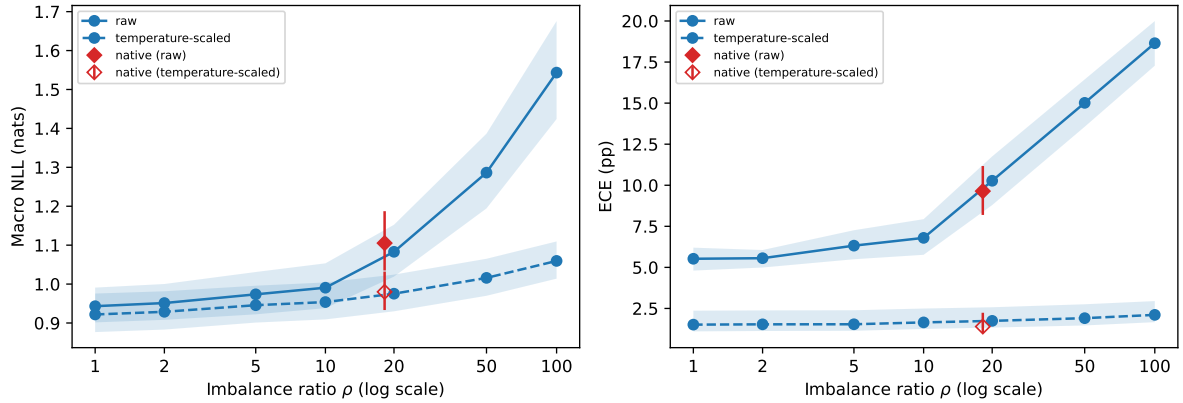


Figure 3: Raw probability quality degrades much faster than accuracy but recovers almost completely under temperature scaling. Left: macro NLL; right: ECE; both raw (solid, filled) and after validation temperature scaling (dashed, open). Blue lines connect the ratio arms at their realized ratio, with shaded paired 95% bootstrap bands; the red diamonds mark the native allocation at its realized ratio of 18.1.

The fitted temperatures show the direction of the miscalibration. The balanced arm is slightly underconfident, with a mean temperature of 0.85, and the temperature rises steadily with ρ to 2.16 at $\rho = 100$, where predictions must be softened by more than a factor of two. The selected regularization moves in the same direction: the balanced arm selects $\lambda = 0.1$ in 27 of 30 fits, whereas $\rho = 100$ selects $\lambda \leq 10^{-3}$ in all 30 fits and the smallest grid value, 10^{-6}, in 2. Refitting these two boundary fits with λ down to 10^{-10} gives a validation balanced accuracy at least 0.05 points lower at every smaller value, so the grid boundary does not constrain the selection. Weaker regularization lets the logits grow, which produces overconfidence. The study does not separate this effect of model selection from the direct effect of the skewed class prior on confidence.

5 Discussion

Of the three readings fixed in Section 3, the results select the second: prevalence alone damages discrimination on TCGA-UT, but only from $\rho = 50$ on, since the damage at $\rho = 20$ falls just short of one point, and by less than three points even at $\rho = 100$. The third reading is not supported, because the native allocation lies on the curve. With patients and the total budget

held fixed, a hundredfold imbalance leaves the tail classes with between one and two patches per patient, and the tail third of the classes still reaches 60 percent recall. Per doubling of the ratio, the loss is small below $\rho = 10$ and grows to 0.65 points above it, which is where tail support per patient falls below eight patches.

For the prevalence factor, the curve replaces the benchmark's three-allocation readout. The benchmark measured 1.33 to 1.79 points of severe-imbalance damage on TCGA-UT (Queisler, 2026b), less than the 2.76 points at $\rho = 100$ here. The benchmark's imbalanced conditions also changed the patient pool and compared against a different reference allocation and budget, so the two magnitudes are not directly comparable; both place the damage of severe prevalence imbalance on TCGA-UT at a few points. What the curve adds is its shape: the benchmark could not show that the damage stays below one point up to $\rho = 20$ and accelerates beyond $\rho = 10$, nor that the native distribution, at a realized ratio of 18, costs just above the one-point threshold. That damage of 1.17 points is also small compared with the patient-count effects of this series: at 160 patches per class, going from five to twenty patients per class gains 9.41 points (Queisler, 2026c). In a long-tailed TCGA-UT training set, the number of patients per class is therefore likely to matter more than the prevalence of the class, provided the rare classes still receive a few patches from each of their patients.

Calibration responds to prevalence much more strongly than discrimination, but in a form that one parameter repairs. The ECE more than triples at $\rho = 100$ while accuracy loses about four percent of its value, and validation temperature scaling on the natural validation distribution removes almost all of that ECE increase. This matches the benchmark's finding that most probability-quality benefits of prevalence weighting disappear after temperature scaling (Queisler, 2026b). For TCGA-UT at fixed patient support, the calibration damage of imbalance is therefore mostly a uniform overconfidence that a post-hoc temperature removes, not a class-specific distortion. Prevalence-based corrections (Buda et al., 2018) could still recover part of the tail recall, but they would have to beat a temperature-scaled baseline on probability quality to show value beyond it.

This study is the first of the studies that split the benchmark into its deprivations. The remaining ones vary patch count per class at fixed prevalence and patient count at fixed patch count, one factor at a time and with the same plain baseline. Together with this curve, they replace the benchmark's bundled conditions and show which of its factors carries the damage it measured.

6 Limitations

Each ratio moves the class prior and the tail support together, because the total budget is fixed; the study does not separate the effect of prevalence as such from the effect of the smaller per-patient patch count of the tail classes. A design at fixed tail support, with more patches for the head, would separate them. The twenty patients per class are the same in every arm, so the curve describes imbalance at adequate patient support; long-tailed datasets usually also have fewer patients in the rare classes, and that combination is left to the patient-shortage study.

The ratio grid ends at 100 and resolves the onset only to the grid spacing: the one-point damage is crossed between $\rho = 20$ and $\rho = 50$, and the damage at $\rho = 20$ misses the threshold by less than 0.01 points, well inside its interval. The native arm is a single allocation, so its agreement with the curve does not show that every class order behaves like a random one. Temperature scaling is fitted on a validation partition with the natural class distribution of TCGA-UT; under a different deployment prior, the calibration results may differ. All results are conditional on frozen Virchow2 features, multinomial logistic regression, TCGA-UT, and three overlapping splits.

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